Applicant: LUO Xiaoyu

Program: MSc Artificial Intelligence and Entrepreneurship, The Hong Kong University of Science and Technology

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PERSONAL STATEMENT

The first time I felt the charm of technology was when I assembled my first computer with my own hands. The process of bringing a pile of cold components to life sparked a strong curiosity in me about the principles of hardware-software collaboration. As my studies progressed, I began to focus on deeper questions: how these hardware-driven systems exhibit intelligence under the influence of algorithms and data. During my undergraduate years, diverse project experiences, such as website building, server operation and maintenance, and function development, allowed me to gradually accumulate experience in programming and system construction. They also showed me that current artificial intelligence is becoming the core driver of technological innovation. When I see technologies like AI tools and smart cars turning human imagination into reality, I am even more eager to stand at the forefront of this transformation. I want not only to participate in the design of intelligent systems as a developer but also, from an entrepreneur's perspective, to turn technology into products and services that truly improve people's lives. The combination of AI and entrepreneurship represents the future direction of technological innovation, and it also carries my original aspiration to create the future.

During my undergraduate studies, I systematically built a knowledge system in information technology and AI, and deepened my understanding through the integration of theory and practice. Through courses like *Introduction to Software Engineering*, and *Software System Design and Architecture*, I not only mastered the overall process of the software development lifecycle but also understood the logic of system architecture design, modular management, and network communication. These courses taught me how to analyze problems from a system perspective, design efficient and scalable solutions, and maintain structured thinking in complex projects. In courses closely related to AI, such as *Data Structures and Algorithms*, and *Artificial Intelligence Foundation*, I gradually developed an algorithm-centered way of thinking and understood the entire process from data input and model training to result output. These courses strengthened my abilities in logical reasoning, computational complexity analysis, and algorithm optimization. They also allowed me to understand the working principles of AI systems from the underlying mechanisms.

Beyond the classroom, I continued to deepen my knowledge through project practice. In a traffic inquiry system developed with classmates, I was responsible for frontend development based on the Vue framework. Faced with a completely unfamiliar framework, I read official documents, analyzed open-source projects, and used AI tools for learning. I quickly mastered its component-based logic and data interaction mechanisms, and successfully completed the development task. This experience made me deeply appreciate the mutual promotion between theory and practice: theory provided me with ideas to solve problems, while practice allowed me to verify and optimize these ideas in real scenarios. To further consolidate my knowledge, I built my own GitHub repository, which systematically records the source code of projects I participated in, study notes, and development insights. This not only helped me form a sustainable learning system but also enabled me to quickly pick up and innovate with new technologies. What's more, during a one-month server operation and maintenance internship, I gained a deep understanding of the integrity and complexity of computer systems. My main task was to write automatic update scripts for server data backup to check whether data had been correctly migrated to a new machine. Initially, I tried to judge if the backup was complete by comparing configuration files, but in practice, I found that backup files would update automatically during migration, making this original approach ineffective. After multiple attempts and debugging, I finally determined the success of the migration by having the script, which itself migrated with the system to the new environment, and triggering the new machine's startup. This process not only queried the cloud provider's API using the machine's ID to get its purchase time but also made me realize that the various modules of a system are not isolated but interconnected, enlightening my thinking--If this purchase time is very recent (e.g., within minutes) compared to the script's execution time, it will confirm the environment shift and execute the automatic configuration update. At the same time, when writing backup scripts for a bastion host, I accidentally deleted configuration files due to an operational error. This mistake made me aware of the importance of permission management and risk control in system design and operation. More importantly, whenever I encountered unsolvable errors and sought help from seniors, they would not give me the answer directly. Instead, they guided me to track the root cause of the problem by recording logs and analyzing execution paths. This problem-oriented solution helped me gradually master methods for independent analysis, iterative improvement, and experience summarization.

This internship experience not only deepened my understanding of system operation and maintenance but also shifted my perspective from engineering to entrepreneurial and management thinking. I realized that technological innovation can only be transformed into sustainable value creation when combined with management mechanisms, risk control, and resource integration. During the process of writing scripts and optimizing workflows, I constantly thought about how to improve efficiency and reduce error rates—this is essentially a form of micro-innovation in production processes and a reflection of entrepreneurial thinking at a micro level. I tried to solve systematic problems with limited resources and create higher organizational efficiency. From this, I gradually understood that entrepreneurship is not just about establishing a company but also about transforming technological potential into practical productivity. This experience equipped me with the ability to identify problems from practice, integrate resources, and drive improvements. It also laid a solid practical foundation for my study of entrepreneurship-related content.

Your university's MSc Artificial Intelligence and Entrepreneurship programme provides an ideal platform for me to achieve this goal. The programme offers a dual training path in AI technology and entrepreneurial management. It not only emphasizes the mastery of core AI technologies like machine learning and deep learning but also highlights the transformation of innovative achievements into commercially valuable entrepreneurial projects. This integration of technology and strategy aligns perfectly with my career plan to grow from a technology developer to an AI innovation leader. The *AI Venture for Entrepreneur* course in the programme is particularly appealing to me. It will guide students to learn how to identify market opportunities, evaluate AI entrepreneurial potential, and transform algorithm prototypes into commercializable products. This course design will help me combine my existing engineering background with entrepreneurial thinking, enabling me to comprehensively consider the implementation of innovative solutions from the perspectives of technical feasibility, market demand, and business models. In addition, the programme cultivates students' leadership and critical thinking through a combination of seminars and practical teaching. This will allow me to form independent judgment and strategic decision-making abilities in a complex, changing technological environment. I believe your programme will not only strengthen my AI technical capabilities but also systematically enhance my comprehensive qualities in innovation management, project incubation, and entrepreneurial execution. It will lay a solid foundation for me to promote technological innovation and social value transformation as an AI entrepreneur in the future.

After systematic academic training and diverse practical experiences, I have a clear plan for my future career direction. After graduation, I plan to join an AI technology-driven enterprise, working on AI system development and innovative product design to accumulate comprehensive experience in technology R&D and project management. Within 3 to 5 years, I hope to establish an innovative startup company centered on AI technology, focusing on applying AI to the intelligent solution of real-world problems—such as urban traffic optimization, enterprise information security, and automated system construction.

实习证明

兹有华南师范大学软件工程专业罗霄羽同学，身份证号为430922200408089613，于2025年7月2日至2025年9月30日在我司进行实习， 其实习岗位是:技术研发部实习生。 该同学在实习期间主要参与了以下工作:

一、协助开发二进制模糊测试agent的开发，负责基础业务框架的构建以及模糊测试过程大语言模型的应用。

二、参与llm驱动符号化漏洞检测器的开发，辅助进行漏洞信息搜集以及通用大模型解析效率的对比实验。

该同学具备积极的学习态度、良好的团队协作能力，并在专业知识的深化理解、实际操作技能及问题解决能力方面取得显著进步。

特此证明。

北京雁翎网卫智能科技有限公司

单位名称：

时间：2025.10.14

Internship Certificate

This is to certify that Mr. LUO Xiaoyu, ID no. 430922200408089613, majoring in Software Engineering at South China Normal University, interned as Technical R&D Intern at our company from July 2, 2025, to September 30, 2025. During the internship, he mainly engaged in the following tasks :

1. Assisted in developing a binary fuzzing test agent, responsible for constructing the foundational business framework and implementing large language model (LLM) applications in the fuzzing process.
2. Contributed to the development of an LLM-driven symbolic vulnerability detector, supporting vulnerability information collection and comparative experiments on parsing efficiency of general-purpose large models.

Mr. LUO demonstrated a proactive learning attitude and strong teamwork skills. Through this internship, he significantly enhanced his technical expertise, practical problem-solving abilities, and operational proficiency.

Hereby Certify.

Company:

(Stamp)

Date: 2025.10.14

广州网律互联网科技有限公司

实习证明

兹有华南师范大学软件工程专业罗霄羽同学，身份证号为430922200408089613，于2024年10月29日至2024年12月12日在我司进行实习，其实习岗位是：技术研发部实习生。该同学在实习期间主要参与了以下工作：

一、协助开发内部坐席管理系统，负责异常处理部分模块的代码编写、测试和优化。

二、参与公司内部软件系统的维护和升级，解决了堡垒机服务器自动化漂移脚本及开机自检脚本编写等技术问题。

纵观他实习期间的表现，他能够积极学习新的计算机技术和知识，与团队成员合作良好，展现出较强的学习能力和团队协作精神。通过实习，该同学对计算机专业知识有了更深入的理解和掌握，具备了一定的实际操作能力和问题解决能力。

特此证明。

公司名字：广州网律互联网科技有限公司

联系电话：020-88801242

公司盖章：

负责人签字：

日期：2024年12月12日

Guangzhou Wanglv Internet Technology Co., Ltd.

Internship Certificate

This is to certify that Mr. LUO Xiaoyu, ID no. 430922200408089613, majoring in Software Engineering at South China Normal University, interned as intern at the Technical R&D Department of our company from October 29, 2024 to December 12, 2024. During the internship, he mainly engaged in the following tasks :

1. Assisted in developing the internal agent management system, and underwent the coding, testing, and optimization of the exception handling module.
2. Participated in maintaining and upgrading internal software systems, resolving technical issues including automated drift scripting for JumpServer and startup self-check script development.

Based on his performance, I found that he could take the initiative to learn new computer technologies and knowledge, and cooperated well with team members, demonstrating strong learning ability and teamwork spirit. Through the internship, Mr. LUO Xiaoyu has gained a deeper understanding and mastery of computer-related knowledge, and acquired certain practical operation skills and problem-solving abilities.

Hereby Certify.

Company: Guangzhou Wanglv Internet Technology Co., Ltd.

Contact Phone: +86 20 3880 1242

Company Seal: 广州网律互联网科技有限公司

Signature of Authorized Person: 陈力 chenli

Date: December 12, 2024

实习证明

兹有华南师范大学软件工程专业罗霄羽同学，身份证号为：430922200408089613，于 2024 年 7 月 1 日至 2024 年 8 月 31 日在我司进行实习， 其实习岗位是：IT 开发助理工程师。 该同学在实习期间主要参与了以下工作：

一、协助开发配置报价系统，负责部分模块的代码编写、测试和优化。

二、参与公司配置报价系统的维护和升级，完成了梯度管理、参数配置、汇率管理等业务模块开发。

纵观他实习期间的表现，他能够积极学习新的计算机技术和知识，与团队成员合作良好，展现出较强的学习能力和团队协作精神。通过实习，该同学对计算机专业知识有了更深入的理解和掌握，具备了一定的实际操作能力和问题解决能力。

特此证明。

单位名称：深圳麦克韦尔科技有限公司

时间：2024 年 8 月 31 日

深圳麦克韦尔科技有限公司 人力资源专用章

Internship Certificate

This is to certify that Mr. LUO Xiaoyu, ID no. 430922200408089613, majoring in Software Engineering at South China Normal University, interned as IT Development Assistant Engineer at our company from July 1, 2024 to August 31, 2024. During the internship, he mainly engaged in the following tasks :

1. Assisted in the development of Configuration quotation system, and took control of writing, testing and optimizing some modules of the code.

2. Participated in the maintenance and upgrade of the company's configuration quotation system, and completed the development of business modules such as gradient management, parameter configuration, and exchange rate management.

Based on his performance during the internship, he could take the initiative to learn new computer technologies and knowledge, and cooperated well with team members, demonstrating strong learning ability and teamwork spirit. Through the internship, Mr. LUO Xiaoyu has gained a deeper understanding and mastery of computer-related knowledge, and acquired certain practical operation skills and problem-solving abilities.

Hereby Certify.

深圳麦克韦尔科技有限公司

人力资源专用章

Company: Smoore International Holdings Limited

Date: August 31, 2024